NAVRACHANA UNIVERSITY

Teaching Plan

Database Management Systems (CS237)

Credit : 3

Faculty

Name	Contact	Office Hours	Office Location
Avani Vasant	avani.vasant@nuv.ac.in		

School

SET

Semester

Spring (Jan to June 2025)

Lecture time & Weedays

SECTION_1	Friday	16:00:00-17:00:00 12:00:00-13:00:00	08-01-2025 TO 12-01-2025 13-01-2025 TO 09-05-2025
	Monday	12:00:00-13:00:00	08-01-2025 TO 09-05-2025
	Wednesday	10:00:00-11:00:00 10:00:00-11:00:00	08-01-2025 TO 12-01-2025 13-01-2025 TO 09-05-2025
SECTION_2	Monday	14:00:00-15:00:00	08-01-2025 TO 09-05-2025
	Wednesday	14:00:00-15:00:00 14:00:00-15:00:00	08-01-2025 TO 12-01-2025 13-01-2025 TO 09-05-2025
	Friday	10:00:00-11:00:00 10:00:00-11:00:00	08-01-2025 TO 12-01-2025 13-01-2025 TO 09-05-2025

Pre-requisite

Co-requisite

Anti-requisite

Course Description

This course introduces students to basic database design and the use of databases in applications. Throughout the course students are introduced to Database and Database Management System, advantages and disadvantages of the different database models, relational model and Structured Query Language (SQL), Understanding constraints, transaction management, ACID properties. It includes the design using various diagrams like ER, Sequence, and Activity. The designed system will produce the 'reports' for specific queries.

Course Abstract

The course covers the fundamental concepts of database design and implementation, relational concepts, data modeling, conceptual and logical database design, use of SQL as a data- definition, manipulation, control language, and database administration.

Course Objectives

• To introduce fundamental concepts of DBMS.
• To design and create the Database model.
• To understand the Normalization techniques to reduce the data Redundancy.
• To Learn the query processing system, recovery techniques and management of transactions in database.
• To empower and expose the students with database implementation using SQL and PL/SQL languages.

Course Outcomes

CO1 To be able to apply analytical skills to map out the conceptual design for a real-life problem and explain the fundamental elements of Database management System.
CO2 To be able to draw the model of Relational Database using Entity Relationship (ER) model
CO3 To be able to Improve the database design by normalization.
CO4 To be able to evaluate the logical design to translate into a specific data model and physical design to meet system storage requirements.
CO5 To be able to evaluate Oracle features and Oracle related products for maintaining the integrity and performance of databases.

Course Typology

Theory

Course Outline

Data and Information: . Introduction to Structured, Semi-structured and unstructured data. Limitations of Manual Data Processing, Advantages of DBMS over traditional computer file-based processing approach, Functions of DBA.
ENTITY-RELATIONSHIP MODEL: Basic concepts, entity, attributes, relationship, Design process, constraints, Keys, weak entity sets, extended E-R features – generalization, specialization, aggregation, Reduction of ER to Relational database.
RELATIONAL MODEL
Structure of relational databases, Domains, Keys - Primary, Secondary and Composite, Types of Constraint, Anomaly in database design, Functional Dependency- definition, type of FDs, closure of FD set, closure of attributes, Equivalence of FD, irreducible set of FD, Decomposition- lossless and Dependency Preservation, Normalization – 1Nf, 2NF, 3NF, BCNF, 4NF and 5NF.
RELATIONAL ALGEBRA AND SQL
Introduction to query language, Fundamental operators and syntax, relational algebra queries. SQL Concepts: Basics of SQL, DDL, DML, DQL, DCL, TCL.
QUERY AND STORAGE STRATEGIES
Query Processing & Query Optimization: Overview, measures of query cost. Storage strategies: Indices, B-trees, hashing.
TRANSACTION MANAGEMENT
Transaction Management: Transaction concepts, properties of transactions, Transaction states, concurrent executions of transactions and related problems, serializability of transactions, testing for serializability, Types of schedules, concurrency control, two-phase locking protocol.
Reference books:
1. Silberschatz, A., Korth, H. F., & Sudarshan, S. (2010). Database system concepts (int'l ed) (6th ed.). McGraw Hill Higher Education.
2. Date, C. J. (1977). An introduction to database systems. Pearson Education.
3. Date, C. J. (2012). Database design and relational theory: Normals forms and all that jazz. O'Reilly Media.
4. Gruber, M. (1990). Understanding SQL.
5. Ivan Bayross. SQL, PL/SQL – The Programming Language of Oracle

Pedagogy

. Teaching Theory and Practice with Active Learning Approach
• Brainstorming
• Stop-and-Think
• Think-Pair-Share
• Exercise Breaks
• Case Study/Group Problem Solving

Expectations from Students

1. Attendance is mandatory as per university norms.
2. Be Regular in attending sessions and reaching sessions on time.
3. To be attentive and participate in classroom discussions.
4. To refrain from using mobile phones in the classroom.
5. Read concepts before the class and revise after the class.
6. Do assigned homework every day; this is the only way to learn and succeed.
7. Spend 5 hours a week reading the course from reference books.
8. Ask for help when appropriate.
9. Maintain academic integrity in assignments.

Assessment / Evaluation

Project / Assignment Details

Will be discussed in classroom and assigned through Teams

Course Material

PPTs, Lecture Notes will be posted progressively.

Additional Information

Students are encouraged
To learn from NPTEL Archive courses on Database Management System

Session Plan

Topic Title	Session No.	Topic & Subtopic Details	Readings, Cases, etc.	Activities	Important Dates
DATA AND INFORMATION	1	Database-System Applications, Purpose of Database Systems, View of Data	Reference Books	Brainstorming	
DATA AND INFORMATION	2	Limitations of Manual Data Processing, Advantages of DBMS over traditional computer file-based processing approach	Reference Books	Discussion	
DATA AND INFORMATION	3	Database Languages, Relational Databases, Database Users and Administrators, Functions of DBA	Reference Books	Learning	
ENTITY-RELATIONSHIP MODEL	4	Basic concepts, entity, attributes, relationship	Reference Books	Think-Pair-Share	
ENTITY-RELATIONSHIP MODEL	5	Design process, constraints, Keys	Reference Books	Learning	
ENTITY-RELATIONSHIP MODEL	6		Reference Books	Brainstorming	
ENTITY-RELATIONSHIP MODEL	7	Reduction of ER to Relational database	Reference Books	Problem Solving	
RELATIONAL MODEL	8	Structure of relational databases, Domains	Reference Books	Active Learning	
RELATIONAL MODEL	9	Keys - Primary, Secondary and Composite	Reference Books	Think-Pair-Share	
RELATIONAL MODEL	10	Types of Constraint	Reference Books	Active Learning	
RELATIONAL MODEL	11	Anomaly in database design, Functional Dependency-definition	Reference Books	Discussion	
RELATIONAL MODEL	12	type of FDs, Decomposition- lossless and Dependency Preservation	Reference Books	Problem Solving	
RELATIONAL MODEL	13		Reference Books	Group Problem Solving	
RELATIONAL MODEL	14		Reference Books	Group Problem Solving	
RELATIONAL MODEL	15		Reference Books	Group Problem Solving	
RELATIONAL MODEL	16		Reference Books	Group Problem Solving	
RELATIONAL MODEL	17		Reference Books	Group Problem Solving	
RELATIONAL MODEL	18		Reference Books	Group Problem Solving	
RELATIONAL MODEL	19	closure of FD set	Reference Books	exercise breaks	
RELATIONAL MODEL	20	closure of attributes	Reference Books	exercise breaks	
RELATIONAL MODEL	21	Equivalence of FD, irreducible set of FD	Reference Books	exercise breaks	
RELATIONAL ALGEBRA AND SQL	22	Introduction to query language, Fundamental operators and syntax	Reference Books	Discussion	
RELATIONAL ALGEBRA AND SQL	23	Introduction to query language, Fundamental operators and syntax	Reference Books	Discussion	
RELATIONAL ALGEBRA AND SQL	24	relational algebra queries	Reference Books	Problem Solving	
RELATIONAL ALGEBRA AND SQL	25	relational algebra queries	Reference Books	Group Problem Solving	
RELATIONAL ALGEBRA AND SQL	26	SQL Concepts: Basics of SQL, DDL, DML, DQL, DCL, TCL	Reference Books	Exercises	
RELATIONAL ALGEBRA AND SQL	27	SQL Concepts: Basics of SQL, DDL, DML, DQL, DCL, TCL	Reference Books	Exercises	
QUERY AND STORAGE STRATEGIES	28	Query Processing & Query Optimization: Overview, measures of query cost	Reference Books	Active Learning / Presentations	
QUERY AND STORAGE STRATEGIES	29	Query Processing & Query Optimization: Overview, measures of query cost	Reference Books	Active Learning / Presentations	
QUERY AND STORAGE STRATEGIES	30	Query Processing & Query Optimization: Overview, measures of query cost	Reference Books	Active Learning / Presentations	
QUERY AND STORAGE STRATEGIES	31	Storage strategies: Indices, B-trees, hashing	Reference Books	Active Learning / Presentations	
QUERY AND STORAGE STRATEGIES	32	Storage strategies: Indices, B-trees, hashing	Reference Books	Active Learning / Presentations	
QUERY AND STORAGE STRATEGIES	33	Storage strategies: Indices, B-trees, hashing	Reference Books	Active Learning / Presentations	
QUERY AND STORAGE STRATEGIES	34	Storage strategies: Indices, B-trees, hashing	Reference Books	Active Learning / Presentations	
TRANSACTION MANAGEMENT	35	Transaction Management: Transaction concepts, properties of transactions	Reference Books	Brainstorming & Discussion	
TRANSACTION MANAGEMENT	36	Transaction Management: Transaction concepts, properties of transactions	Reference Books	Brainstorming & Discussion	
TRANSACTION MANAGEMENT	37	Transcation states, Schedules	Reference Books	Exercises	
TRANSACTION MANAGEMENT	38	concurrent executions of transactions and related problems	Reference Books	Case study/ Group Problem Solving	
TRANSACTION MANAGEMENT	39	concurrent executions of transactions and related problems	Reference Books	Case study/ Group Problem Solving	
TRANSACTION MANAGEMENT	40	serializability of transactions	Reference Books	Case study/ Group Problem Solving	
TRANSACTION MANAGEMENT	41	testing for serializability	Reference Books	Brainstorming & Discussion	
TRANSACTION MANAGEMENT	42	Types of schedule	Reference Books	Exercises	
TRANSACTION MANAGEMENT	43	concurrency control	Reference Books	Exercises	
TRANSACTION MANAGEMENT	44	two-phase locking protocol	Reference Books	Think-Pair-Share	
Future Directions & Reflections	45	Emerging Trends, Research, or Advancements in the field of Database Management, Personal Insights and Learning Outcome		Discussions/ Sharing/ Group Sharing	

Publish Status	Published